

WORKFORCE LENS

See the workforce clearly. Act before role change.

90-Day Product Strategy for Medika Nusantara

A decision product for reskilling and role transition - built on a trustworthy workforce data layer.

Deliverables included:

1. Product and data strategy
2. Working prototype
3. Synthetic datasets

A Product Manager Case Study Submission for Rakamin

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Executive position

Do not build an attrition predictor first. Build a reskilling and role-transition decision product.

The CHRO does not need “AI workforce insights.” She needs a defensible way to decide which employee groups should be assessed, reskilled, or moved before targeted roles change - without replacing source systems and without payroll data. The 90-day PoC therefore proves two things: (1) a contextualized data layer can convert fragmented HR records into decision-grade evidence, and (2) HR managers can act through a prioritized validation queue rather than a passive dashboard.

Assumption	Why it is explicit
Scope unit	The PoC uses a representative 200-person sample across role families and branches; production rollout would scale to all 15,000 employees.
Legal boundary	No Oracle payroll data, payroll-derived proxy, or compensation inference is used.
Outcome definition	A “successful decision” is a validated reskilling or transition action, not a model score.
Ground truth	Manager validation, course completion, skill assessment and mobility outcomes become the feedback loop; historical labels alone are insufficient.
Role exposure	Two-year role exposure is initially expert-configured and later calibrated with business process and technology-roadmap evidence.

1. Problem discovery

1.1 Actual product problem

Product problem: HR leaders cannot reliably identify and validate employees for role-transition interventions because workforce evidence is fragmented, non-comparable, and incomplete. The job is not to generate predictions; it is to reduce the time and risk involved in moving from evidence to a reskilling decision.

- **Primary user:** Regional/branch HR manager accountable for workforce actions, supported by HR data stewards and functional managers.
- **Job to be done:** “When a role family is exposed to change, help me identify who should be assessed first, understand why, and initiate the right validation or development action.”
- **Success metric:** Validated action rate: percentage of recommendations that become an agreed assessment, learning plan, mobility conversation, or data-correction task within 30 days.
- **Guardrail:** No adverse employment action may be taken solely from a model recommendation.

1.2 Focus-area trade-off

Focus area	Data availability	Business impact	90-day feasibility	PoC speed	Decision
Attrition prediction	Low: payroll unavailable; historical performance unreliable	High, but errors create employee-relations risk	Low-medium	Medium	Defer
Skill-gap + reskilling	Medium: LMS is messy but recoverable;	High and directly tied to transformation	High	High	Prioritize

	role/title signals available				
Internal mobility matching	Medium: IDs and skills must be resolved first	High; converts reskilling into placement	Medium	Medium	Design next
Strategic workforce planning	Medium-low: demand scenarios and role forecasts not yet validated	Very high at executive level	Low for individual decisions	Low	Use as aggregate lens, not PoC core

Choice: skill-gap and role-transition prioritization wins because it creates an actionable output using recoverable evidence, demonstrates the value of contextualization, and establishes reusable primitives for later mobility and workforce-planning products.

1.3 First two weeks: critical learning agenda

Critical question	How I would answer it
Which decision will the CHRO defend at Day 90?	Run a 90-minute decision-backwards workshop with CHRO, HRBPs, L&D and transformation lead; force a single decision, owner, cadence and success criterion.
Which role families are genuinely changing?	Interview operations, sales, IT and transformation leaders; map process/technology changes to 3-5 exposed role families and obtain named executive sponsors.
What data may legally be processed?	Complete a field-level data inventory and privacy review; classify direct identifiers, sensitive attributes, allowed purposes, retention and access roles. Confirm payroll exclusion in writing.
Can identities be resolved without payroll?	Profile SuccessFactors, Workable and Moodle keys; sample 500 records; test deterministic and probabilistic match rates; quantify ambiguous and unmatched cases.
What does “skill” mean locally?	Facilitate role-to-skill calibration with L&D and 8-12 managers; reconcile Moodle categories with a configurable taxonomy and proficiency scale.
Which evidence do managers trust?	Shadow 5 HR managers reviewing employee cases; test explanation formats, confidence language and required drill-down evidence.
What outcome can become ground truth?	Define feedback events: manager agrees/disagrees, assessment result, course completion, transition accepted, and outcome after 90/180 days.

1.4 Explicit 90-day non-goals

- Individual attrition prediction or automated “flight risk” labeling: labels and features are too weak, and payroll is unavailable.
- Company-wide role obsolescence forecast for every job: initially constrain to 3-5 sponsored role families.
- Autonomous employee decisions: every recommendation is decision support and requires manager validation.
- Replacing or rebuilding HRIS/ATS/LMS: use connectors and a canonical layer above them.
- Reconstructing pre-2022 performance history: quarantine it unless independently validated.
- Production-grade MLOps for all 15,000 employees: build a thin, auditable vertical slice and a scaling design.

2. Data strategy

2.1 From fragmented records to usable evidence

1. Contract - Field-level source contracts, allowed purpose, owner, refresh cadence, quality tests and legal classification.

2. Profile - Measure completeness, uniqueness, scale drift, missingness patterns and branch-specific semantics.

- 3. **Resolve identity** - Create a crosswalk and canonical employee ID using deterministic rules first, then probabilistic matching.
- 4. **Standardize** - Normalize titles, branches, dates, rating scales and skill vocabulary; retain raw values and lineage.
- 5. **Enrich carefully** - Infer skills only from approved evidence such as title taxonomy, completed courses and assessments; attach provenance and confidence.
- 6. **Serve** - Publish AI-ready employee-role-skill snapshots plus a review queue; never overwrite systems of record.
- 7. **Learn** - Capture manager validation and downstream actions as new labeled evidence.

2.2 Logical architecture

Layer	Responsibility	Key controls
Sources	SuccessFactors, Workable, Moodle, custom performance tool	Read-only connectors; payroll excluded
Raw landing	Immutable source snapshots	Encryption, source timestamp, schema version, access logging
Quality + observability	Completeness, freshness, uniqueness, scale and drift tests	Fail/quarantine thresholds; branch-level scorecards
Identity graph	Cross-system person entities and source-ID crosswalk	Deterministic/probabilistic evidence, match score, reversible merges
Canonical workforce model	Employee, position, role, skill, learning, performance event	Bitemporal history, raw-to-canonical lineage
Contextualization services	Title normalization, skill taxonomy mapping, rating-scale normalization, inference	Model/rule version, confidence, provenance
Feature + decision layer	Point-in-time features and reskill recommendations	No leakage, eligibility rules, reason codes, confidence calibration
Experience layer	Decision queue, evidence drawer, review workflow	Role-based access, human approval, feedback capture

2.3 Identity resolution without Oracle

Treat identity resolution as an evidence graph, not a one-time join. Oracle ID MN-444 is unavailable and therefore not required for the PoC. The canonical crosswalk links only approved source IDs and can later add payroll as a separately governed edge if policy changes.

- **Deterministic pass:** Exact normalized work email; approved HRIS-to-LMS mapping; exact employee number embedded in an allowed field; exact name + branch + start date when unique.
- **Probabilistic pass:** Weighted similarity on normalized name, email local-part, phone hash if allowed, branch, role family and start-date proximity. Use Fellegi-Sunter-style evidence or an explainable gradient model; do not use a black-box LLM for final linkage.
- **Decision thresholds:** ≥0.92 auto-link; 0.75-0.91 human review; <0.75 remain unlinked. Conflicting high-trust identifiers block auto-merge.
- **Merge safety:** Preserve every source record, match evidence, rule/model version and reviewer decision. Support split/undo.
- **Quality metric:** Precision on auto-links is the primary guardrail; target ≥99%. Recall can improve gradually through review.

3. AI system design

3.1 90-day component: Reskill Transition Recommender

The PoC ranks employees within an approved role family for a specific target-role pathway. It does not decide who is “good” or “bad”; it recommends who should enter a manager validation conversation first.

Feature family	Examples	Use
Role exposure	Expert-configured 24-month exposure by canonical role; process automation roadmap	Creates urgency, not an individual judgment
Skill evidence	Observed LMS skills, completed learning, assessments, approved role-based inferred skills	Estimates target-role gap
Trajectory	Normalized 2022-2025 performance trend and consistency	Indicates capacity to absorb transition; missingness remains explicit
Engagement proxy	Allowed survey/engagement score where available	Weak supporting feature, never sole trigger
Mobility constraints	Branch, role eligibility, tenure bands, manager validation	Filters infeasible recommendations
Data quality	Identity, skill, performance and record confidence	Calibrates recommendation confidence and review routing

Output per employee: target role, reskill-priority score, transition-readiness score, top evidence/reason codes, missing-evidence warnings, record confidence, recommendation confidence, and next-best action (manager validate, assess skills, correct data, or create learning plan).

3.2 Missing data strategy

- Distinguish unknown from absent. A blank skill field never means “no skill.”
- Impute only for model continuity; never display imputed values as observed facts.
- Infer candidate skills from normalized role, completed Moodle courses, assessment evidence and manager-validated equivalents. Store source, method, timestamp and confidence for every inferred skill.
- Use missingness indicators as features because branch/system adoption patterns are informative, while auditing them to avoid penalizing poorly digitized branches.
- Cap recommendation confidence when critical evidence is inferred; route low-confidence high-impact cases to review.
- Create an active-learning queue: ask managers the smallest number of questions that most improves decision confidence.

3.3 Confidence signaling

Band	User sees	System behavior
High ($\geq 85\%$)	“Ready to discuss” with 2-3 reason codes and evidence links	May enter manager queue; still requires human confirmation
Medium (70-84%)	“Validate evidence” plus exactly what is inferred or missing	Prompt targeted skill assessment or manager confirmation
Low ($< 70\%$)	“Do not act - data review required”	No recommendation; create steward task
Human-review override	Conflicting identity, inferred critical skill, out-of-distribution role/branch, sharp model-manager disagreement, or protected-attribute proxy concern	Block automation and log review outcome

3.4 Day 75: “82% accuracy, but it feels wrong”

I would stop the launch decision. Accuracy is not sufficient evidence of usefulness and may hide class imbalance, label leakage, branch bias or a mismatched outcome definition.

1. Reconstruct the metric: define label, base rate, split strategy, time leakage and comparison baseline. Report precision/recall, calibration and error cost by branch/role - not only accuracy.
2. Run structured error review with the three managers. For each disputed case, separate data error, explanation failure, policy disagreement and legitimate model discovery.
3. Check feature and label integrity: rating-scale normalization, missingness effects, duplicate leakage, post-outcome features, and proxy variables.

- 4. Compare against simple baselines and manager predictions. A complex model must beat role rules and manager-only judgment on the chosen decision metric.
- 5. Recalibrate or reframe. If ground truth is weak, downgrade from prediction to risk triage and collect prospective validation labels.
- 6. Launch only as a monitored shadow workflow until agreement, calibration and intervention usefulness meet predefined gates.

4. Product and UX design

4.1 Decision surface, not dashboard

Surface	What the HR manager needs
Transformation lens	Select exposed role family, target pathway, branch and decision deadline.
Prioritized queue	Employee, current→target role, priority, confidence band, reason summary and required action.
Evidence drawer	Observed vs inferred skills, normalized performance trend, identity quality, source lineage and missing evidence.
Action panel	Create manager validation task, request skill assessment, propose learning pathway, mark not suitable, or send to data steward.
Team summary	Counts of ready/validate/review, expected skill gaps and intervention capacity - no vanity charts.
Feedback capture	One-click agree/disagree plus reason, resulting action and later outcome.

4.2 Communicating uncertainty without eroding trust

Show uncertainty as an operational next step, not as statistical decoration. Use three plain-language states: Ready to discuss, Validate evidence, and Data review required. Always pair confidence with its cause (“skills inferred from role taxonomy,” “start date conflicts,” “performance history missing”) and the action that resolves it. Avoid decimal probabilities for consequential people decisions.

5. 90-day roadmap: every milestone is demo-able

Day	C-suite demo	Product learning / exit criteria
0-14	Decision contract: one exposed role family, one target pathway, named user, success metric, privacy boundary and sample evidence cards	CHRO signs the decision scope; legal confirms no payroll; managers agree on review language
30	Data Trust Console: source profile, canonical schema, title taxonomy v1, first identity graph, and before/after records for 50 employees	≥99% precision on auto-linked validation sample; data gaps and owners are visible
60	Working decision queue for 200 employees: normalized titles, skill inference with provenance, confidence bands, and manager feedback workflow	≥70% of sampled recommendations judged plausible enough to validate; critical errors categorized

75	Shadow evaluation: model/baseline comparison, calibration, branch/role error slices, disagreement review	No unresolved leakage; recommendation confidence calibrated; launch gate decision
90	C-suite decision pilot: one role-family transition portfolio, named interventions, review backlog, value-of-data comparison and scale plan	Managers create real actions; CHRO sees how contextualization changes decisions and approves next rollout

6. From custom engagement to reusable platform

Productize the invariants; configure the client-specific semantics. The platform is not one universal model. It is a governed assembly system for workforce evidence and decision workflows.

Reusable across clients	Configurable per client	Never silently customized
Connector framework and raw contracts	Source-field mappings and refresh cadence	One-off SQL embedded in production
Canonical employee-position-role-skill-event model	Organization/branch hierarchy and local role aliases	Client-specific tables without schema ownership
Identity resolution service + review UI	Match thresholds and allowed identifiers	Irreversible merges
Taxonomy service with versioning	Role-to-skill mappings, proficiency, language aliases	Taxonomy changes without lineage
Data-quality rules and scorecards	Thresholds and accountable owners	Missingness treated as absence
Modular AI services: normalize, infer, match, rank, explain, calibrate	Use-case policy, role exposure, eligibility, interventions	A monolithic client-trained model
Decision workflow and feedback event schema	Approval chain and HR operating model	Scores without action capture
Evaluation harness and model cards	Business metric and error-cost weights	Accuracy-only launch gates

6.1 What the next five clients reuse

- 70-80% of the technical spine: connectors, canonical model, identity graph, quality tests, taxonomy tooling, confidence framework, evaluation harness and decision components.
- 40-60% of the semantic layer: common healthcare/distribution roles and skills can seed configuration, but must be locally validated.
- 100% of feedback-event definitions so every engagement compounds Rakamin's evidence about which recommendations and interventions work.
- No raw client data is pooled by default. Cross-client learning uses permissioned, de-identified patterns, reusable ontology assets and federated/aggregate insights under explicit contracts.

7. Synthetic data and value-of-data demonstration

Dataset A contains 220 source-like rows representing 200 employees, including 20 duplicate-ish records. 47% of primary records have blank skill fields, titles vary by branch, performance scales and history are inconsistent, IDs differ, and some dates conflict. Dataset B contains 200 canonical employees with normalized titles, resolved IDs, observed/inferred skill provenance, field confidence and human-review flags.

Measure	Messy input	Contextualized input	Why it matters
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Skill completeness	53%	100% populated; inferred fields labeled	Unknown is separated from absent
Actionable recommendations	Confidence-capped; many cases blocked	186 high-confidence records; 6 review flags	HR receives a safe queue, not a false ranking
Top-40 overlap with contextualized priority	60%	Reference decision set	Raw output wrongly flags or misses employees because aliases and blanks distort features
Identity	System-specific IDs + duplicate-ish rows	Canonical ID + crosswalk + reversible evidence	Prevents duplicate intervention and leakage
Explanation	Raw fields with unclear semantics	Reason codes, provenance, confidence and next action	Builds trust through inspectability

7.1 Skill inference logic

For records with observed skills, values are normalized to the canonical vocabulary. For blank skill fields, Dataset B proposes three candidate skills from the validated canonical role template. In production, completed courses, assessments and manager evidence would strengthen or reject each candidate. Role-only inference is capped at 68% skill confidence and remains visibly marked; it is never represented as a verified employee capability.

8. Measurement, risk and governance

Category	Metric / gate
Decision value	Validated action rate; time from exposed role selection to intervention; manager adoption
Recommendation quality	Precision@K for manager-validated candidates; calibration; disagreement resolution rate
Data layer	Identity auto-link precision; canonical completeness; unresolved conflicts; lineage coverage
Fairness	Selection and error-rate slices by branch, tenure band, gender/age only where legally allowed and solely for audit; never use protected attributes to rank
Safety	Zero automated adverse decisions; 100% low-confidence cases blocked; access and review logs complete
Platform	Configuration vs custom-code ratio; connector reuse; taxonomy reuse; onboarding time for next client

9. Deliverable index and usage

Live prototype - Workforce Lens: <https://miftaadiwira16.github.io/workforce-lens/>

Synthetic datasets and source repository: <https://github.com/miftaadiwira16/workforce-lens>

medika_messy.csv / .json - source-like enterprise reality; medika_clean.csv / .json - canonical post-contextualization records.

README.md - product overview, data assumptions, and GitHub Pages deployment notes.

10. AI tools and working method

I used generative AI as a structured copilot to: decompose the case requirements; pressure-test scope choices; generate and validate synthetic edge cases; draft the standalone HTML/CSS/JavaScript prototype; and improve document clarity. I retained product judgment for the core decisions, explicitly constrained the synthetic-data logic, inspected generated outputs, and used

deterministic scripts for reproducibility. In a real engagement, no production employee data would be entered into a public AI tool without an approved enterprise environment, data-processing agreement and privacy review.

Least-confident decision

The decision I am least confident about is using role exposure as a meaningful 24-month signal in the first PoC. It is directionally aligned with the CEO's transformation mandate, but it can become opinion dressed as data if the process and technology roadmap is vague. I would validate it through a structured role-impact workshop with business leaders, process owners and employees, then compare expert exposure ratings with observed task changes and external role evidence. Until that validation is complete, exposure remains a configurable scenario input - not a learned fact.

Appendix A - Dataset data dictionary

Messy dataset

- source_row_id
- hris_employee_id
- ats_candidate_id
- lms_user_id
- employee_name
- email
- branch
- job_title_raw
- hris_start_date
- ats_start_date
- skills_raw
- performance_history_raw
- engagement_score
- record_source
- known_conflict

Clean dataset

- canonical_employee_id
- employee_name
- branch
- hris_employee_id
- ats_candidate_id
- lms_user_id
- normalized_job_title
- target_role
- canonical_start_date
- skills
- skill_source
- inferred_skills
- performance_history
- engagement_score
- role_obsolescence_risk

- skill_gap_score
- reskill_priority_score
- transition_readiness_score
- identity_confidence
- skill_confidence
- record_confidence
- human_review_flag
- review_reason